

North Carolina A&T State University

Department of Mechanical Engineering

MEEN 424

Vibrations & Control

Session 3 Lecture Notes — Part 2

Inverse Laplace Transforms & Transfer Functions

Chapter 2 — Sections 2.3–2.4

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Instructor: Amanuel Abrdo Tereda

Email: aatereda@ncat.edu

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These notes continue directly from Session 3 Part 1 (Laplace Transform Review).
They supplement, but do not replace, attendance and active participation.

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1. Section 2.3 (continued): Inverse Laplace Transforms

1.1. Definition and Notation

Given a Laplace-domain function $F(s)$, we want to recover the original time-domain function $f(t)$. This is called the **Inverse Laplace Transform** and is written:

$$f(t) = \mathcal{L}^{-1}\{F(s)\}$$

1.2. Linearity of the Inverse Transform

Just as the forward transform is linear, so is the inverse:

Linearity Property: Given two Laplace transforms $F(s)$ and $G(s)$, and any constants a and b :

$$\mathcal{L}^{-1}\{aF(s) + bG(s)\} = a\mathcal{L}^{-1}\{F(s)\} + b\mathcal{L}^{-1}\{G(s)\}$$

This means we can split $F(s)$ into simpler terms, invert each one using the table, and add the results.

1.3. General Strategy for Inverse Transforms

The standard approach for finding $f(t) = \mathcal{L}^{-1}\{F(s)\}$ is:

1. **Partial fraction decomposition** — break $F(s)$ into a sum of simple terms
2. **Match each term** to an entry in the Laplace table
3. **Fix numerators** if needed (multiply and divide by a constant)
4. **Apply the inverse transform** term by term using linearity

1.4. Relevant Table Entries for Inverse Transforms

The following entries from the standard Laplace table are used most frequently when computing inverse transforms. Each entry is presented as $F(s) \rightarrow f(t)$.

Entry	$F(s)$	$f(t) = \mathcal{L}^{-1}\{F(s)\}$
#1	$\frac{1}{s}$	1
#2	$\frac{1}{s-a}$	e^{at}
#3	$\frac{n!}{s^{n+1}}$	$t^n, \quad n = 1, 2, 3, \dots$
#7	$\frac{a}{s^2 + a^2}$	$\sin(at)$
#8	$\frac{s}{s^2 + a^2}$	$\cos(at)$
#17	$\frac{a}{s^2 - a^2}$	$\sinh(at)$
#18	$\frac{s}{s^2 - a^2}$	$\cosh(at)$

2. Worked Examples: Inverse Laplace Transforms

Problem: Find $f(t)$ given

$$F(s) = \frac{6}{s} - \frac{1}{s-8} + \frac{4}{s-3}$$

Relevant table entries:

$$\mathcal{L}^{-1}\left\{\frac{1}{s}\right\} = 1, \quad \mathcal{L}^{-1}\left\{\frac{1}{s-a}\right\} = e^{at}$$

Solution: Apply linearity directly — each term already matches entry #1 or #2:

$$\begin{aligned} f(t) &= 6(1) - e^{8t} + 4e^{3t} \\ &= \boxed{6 - e^{8t} + 4e^{3t}} \end{aligned}$$

Problem: Find $h(t)$ given

$$H(s) = \frac{19}{s+2} - \frac{1}{3s-5} + \frac{7}{s^5}$$

Relevant table entries:

$$\mathcal{L}^{-1}\left\{\frac{1}{s-a}\right\} = e^{at}, \quad \mathcal{L}^{-1}\left\{\frac{n!}{s^{n+1}}\right\} = t^n$$

Solution:

Term 1: $\frac{19}{s+2} = \frac{19}{s-(-2)} \rightarrow$ entry #2 with $a = -2$.

Term 2: Factor 3 from denominator: $\frac{1}{3s-5} = \frac{1}{3} \cdot \frac{1}{s-5/3} \rightarrow$ entry #2 with $a = \frac{5}{3}$.

Term 3: Need $4! = 24$ in the numerator for entry #3 ($n = 4$): $\frac{7}{s^5} = \frac{7}{4!} \cdot \frac{4!}{s^{4+1}}$.

Therefore:

$$h(t) = 19e^{-2t} - \frac{1}{3}e^{5t/3} + \frac{7}{24}t^4$$

Problem: Find $f(t)$ given

$$F(s) = \frac{6s}{s^2+25} + \frac{3}{s^2+25}$$

Relevant table entries:

$$\mathcal{L}^{-1}\left\{\frac{s}{s^2+a^2}\right\} = \cos(at), \quad \mathcal{L}^{-1}\left\{\frac{a}{s^2+a^2}\right\} = \sin(at)$$

Solution: Both terms share the denominator $s^2+25 = s^2+5^2$, so $a = 5$.

Term 1: Matches entry #8 directly: $\frac{6s}{s^2+5^2} \rightarrow 6\cos(5t)$.

Term 2: Need a 5 in numerator for entry #7. Multiply and divide: $\frac{3}{s^2+25} = \frac{3}{5} \cdot \frac{5}{s^2+5^2}$.

Therefore:

$$f(t) = 6\cos(5t) + \frac{3}{5}\sin(5t)$$

Problem: Find $g(t)$ given

$$G(s) = \frac{8}{3s^2 + 12} + \frac{3}{s^2 - 49}$$

Relevant table entries:

$$\mathcal{L}^{-1}\left\{\frac{a}{s^2 + a^2}\right\} = \sin(at), \quad \mathcal{L}^{-1}\left\{\frac{a}{s^2 - a^2}\right\} = \sinh(at)$$

Solution:

Term 1: Factor 3 from denominator: $\frac{8}{3s^2 + 12} = \frac{1}{3} \cdot \frac{8}{s^2 + 4} = \frac{1}{3} \cdot \frac{(4)(2)}{s^2 + 2^2}$. Entry #7, $a = 2$:

$$\mathcal{L}^{-1}\left\{\frac{8}{3s^2 + 12}\right\} = \frac{4}{3} \sin(2t)$$

Term 2: $s^2 - 49 = s^2 - 7^2$. Need 7 in numerator: $\frac{3}{s^2 - 49} = \frac{3}{7} \cdot \frac{7}{s^2 - 7^2}$. Entry #17, $a = 7$:

$$\mathcal{L}^{-1}\left\{\frac{3}{s^2 - 49}\right\} = \frac{3}{7} \sinh(7t)$$

Therefore:

$$g(t) = \frac{4}{3} \sin(2t) + \frac{3}{7} \sinh(7t)$$

3. Partial Fraction Decomposition

When $F(s)$ is a ratio of polynomials that does not match a table entry directly, we expand it into simpler fractions. The form of each partial fraction depends on the factors in the denominator.

3.1. Partial Fraction Forms

Factor in denominator	Partial fraction term(s)
$as + b$	$\frac{A}{as + b}$
$(as + b)^k$	$\frac{A_1}{as + b} + \frac{A_2}{(as + b)^2} + \cdots + \frac{A_k}{(as + b)^k}$
$as^2 + bs + c$	$\frac{As + B}{as^2 + bs + c}$
$(as^2 + bs + c)^k$	$\frac{A_1s + B_1}{as^2 + bs + c} + \frac{A_2s + B_2}{(as^2 + bs + c)^2} + \cdots + \frac{A_ks + B_k}{(as^2 + bs + c)^k}$

3.2. Worked Example: Partial Fractions with Linear Factors

Problem: Find $g(t)$ given

$$G(s) = \frac{86s - 78}{(s + 3)(s - 4)(5s - 1)}$$

Step 1 — Write partial fraction decomposition:

$$G(s) = \frac{A}{s + 3} + \frac{B}{s - 4} + \frac{C}{5s - 1}$$

Step 2 — Set numerators equal:

$$86s - 78 = A(s - 4)(5s - 1) + B(s + 3)(5s - 1) + C(s + 3)(s - 4)$$

Step 3 — Solve by substituting roots:

$$s = -3: \quad -336 = A(-7)(-16) \Rightarrow A = -3$$

$$s = \frac{1}{5}: \quad -\frac{304}{5} = C\left(\frac{16}{5}\right)\left(-\frac{19}{5}\right) \Rightarrow C = 5$$

$$s = 4: \quad 266 = B(7)(19) \Rightarrow B = 2$$

Step 4 — Partial fraction result:

$$G(s) = -\frac{3}{s + 3} + \frac{2}{s - 4} + \frac{5}{5s - 1} = -\frac{3}{s + 3} + \frac{2}{s - 4} + \frac{1}{s - \frac{1}{5}}$$

Step 5 — Inverse transform (entry #2, $\mathcal{L}^{-1}\{1/(s - a)\} = e^{at}$):

$$g(t) = -3e^{-3t} + 2e^{4t} + e^{t/5}$$

3.3. Worked Example: Partial Fractions with a Quadratic Factor

Problem: Find $f(t)$ given

$$F(s) = \frac{2 - 5s}{(s - 6)(s^2 + 11)}$$

Step 1 — Decomposition form (linear + irreducible quadratic):

$$F(s) = \frac{A}{s - 6} + \frac{Bs + C}{s^2 + 11}$$

Step 2 — Set numerators equal:

$$2 - 5s = A(s^2 + 11) + (Bs + C)(s - 6)$$

Step 3 — Expand and collect by power of s :

$$2 - 5s = (A + B)s^2 + (-6B + C)s + (11A - 6C)$$

Matching coefficients:

$$A + B = 0, \quad -6B + C = -5, \quad 11A - 6C = 2$$

$$\text{Solving: } A = -\frac{28}{47}, \quad B = \frac{28}{47}, \quad C = -\frac{67}{47}.$$

Step 4 — Factor out $\frac{1}{47}$:

$$F(s) = \frac{1}{47} \left(\frac{-28}{s - 6} + \frac{28s - 67}{s^2 + 11} \right) = \frac{1}{47} \left(\frac{-28}{s - 6} + \frac{28s}{s^2 + 11} - \frac{67}{s^2 + 11} \right)$$

Step 5 — Fix the sine term (need $\sqrt{11}$ in numerator; $a = \sqrt{11}$):

$$\frac{67}{s^2 + 11} = \frac{67}{\sqrt{11}} \cdot \frac{\sqrt{11}}{s^2 + 11}$$

Step 6 — Inverse transform:

$$f(t) = \frac{1}{47} \left(-28e^{6t} + 28 \cos(\sqrt{11}t) - \frac{67}{\sqrt{11}} \sin(\sqrt{11}t) \right)$$

4. Section 2.3: The Transfer Function

4.1. Motivation

Consider a general n th-order, linear, time-invariant (LTI) differential equation relating the output $c(t)$ to the input $r(t)$:

$$a_n \frac{d^n c}{dt^n} + a_{n-1} \frac{d^{n-1} c}{dt^{n-1}} + \cdots + a_0 c(t) = b_m \frac{d^m r}{dt^m} + b_{m-1} \frac{d^{m-1} r}{dt^{m-1}} + \cdots + b_0 r(t) \quad (1)$$

Taking the Laplace transform of both sides and assuming **zero initial conditions**:

$$(a_n s^n + a_{n-1} s^{n-1} + \cdots + a_0) C(s) = (b_m s^m + b_{m-1} s^{m-1} + \cdots + b_0) R(s) \quad (2)$$

4.2. Definition of Transfer Function

Forming the ratio of output to input gives the **transfer function**:

$$G(s) = \frac{C(s)}{R(s)} = \frac{b_m s^m + b_{m-1} s^{m-1} + \cdots + b_0}{a_n s^n + a_{n-1} s^{n-1} + \cdots + a_0}$$

Key points:

- The transfer function *separates* the input, system, and output into distinct algebraic entities.
- It is evaluated with **zero initial conditions**.
- The denominator is identical to the characteristic polynomial of the ODE.
- We can find the output simply as:

$$C(s) = R(s) G(s)$$

- The system is represented as a block diagram: $R(s) \longrightarrow \boxed{G(s)} \longrightarrow C(s)$.

5. System Response from the Transfer Function

Problem: A system has transfer function $G(s) = \frac{1}{s+2}$. Find the response $c(t)$ to a unit step input $r(t) = u(t)$, assuming zero initial conditions.

Relevant facts:

$$r(t) = u(t) \Rightarrow R(s) = \frac{1}{s} \quad (\text{table entry \#1})$$

Step 1 — Form $C(s)$:

$$C(s) = R(s)G(s) = \frac{1}{s} \cdot \frac{1}{s+2} = \frac{1}{s(s+2)}$$

Step 2 — Partial fraction decomposition:

$$\frac{1}{s(s+2)} = \frac{A}{s} + \frac{B}{s+2}$$

Multiplying both sides by $s(s+2)$: $1 = A(s+2) + Bs$.

$s = 0$: $A = \frac{1}{2}$. $s = -2$: $B = -\frac{1}{2}$.

$$C(s) = \frac{1/2}{s} - \frac{1/2}{s+2}$$

Step 3 — Inverse transform:

$$c(t) = \frac{1}{2} - \frac{1}{2}e^{-2t}$$

6. Skill-Assessment Exercises

Skill-Assessment Exercise 2.3

Problem: Find the transfer function $G(s) = C(s)/R(s)$ for the differential equation:

$$\frac{d^3c}{dt^3} + 3\frac{d^2c}{dt^2} + 7\frac{dc}{dt} + 5c = \frac{d^2r}{dt^2} + 4\frac{dr}{dt} + 3r$$

Method: Take the Laplace transform (zero initial conditions), collect $C(s)$ and $R(s)$, form the ratio.

$$(s^3 + 3s^2 + 7s + 5) C(s) = (s^2 + 4s + 3) R(s)$$

Answer:

$$G(s) = \frac{C(s)}{R(s)} = \frac{s^2 + 4s + 3}{s^3 + 3s^2 + 7s + 5}$$

Skill-Assessment Exercise 2.4

Problem: Find the differential equation corresponding to the transfer function:

$$G(s) = \frac{2s + 1}{s^2 + 6s + 2}$$

Method: Cross-multiply: $(s^2 + 6s + 2)C(s) = (2s + 1)R(s)$, then take the inverse Laplace transform.

Answer:

$$\frac{d^2c}{dt^2} + 6\frac{dc}{dt} + 2c = 2\frac{dr}{dt} + r$$

Skill-Assessment Exercise 2.5

Problem: Find the ramp response for a system with transfer function:

$$G(s) = \frac{s}{(s + 4)(s + 8)}$$

Setup: A ramp input is $r(t) = t u(t)$, so $R(s) = \frac{1}{s^2}$.

$$C(s) = R(s) G(s) = \frac{1}{s^2} \cdot \frac{s}{(s + 4)(s + 8)} = \frac{1}{s(s + 4)(s + 8)}$$

Partial fractions:

$$C(s) = \frac{A}{s} + \frac{B}{s + 4} + \frac{C}{s + 8}$$

$$s = 0: A = \frac{1}{32}. \quad s = -4: B = \frac{-1}{(-4)(4)} = -\frac{1}{16}. \quad s = -8: C = \frac{1}{(-8)(-4)} = \frac{1}{32}.$$

Answer:

$$c(t) = \frac{1}{32} - \frac{1}{16}e^{-4t} + \frac{1}{32}e^{-8t}$$

Key Formulas Summary

Concept	Formula
Inverse Laplace	$f(t) = \mathcal{L}^{-1}\{F(s)\}$
Linearity (inverse)	$\mathcal{L}^{-1}\{aF + bG\} = a \mathcal{L}^{-1}\{F\} + b \mathcal{L}^{-1}\{G\}$
Unit step	$\mathcal{L}^{-1}\left\{\frac{1}{s}\right\} = 1$
Exponential	$\mathcal{L}^{-1}\left\{\frac{1}{s-a}\right\} = e^{at}$
Power of t	$\mathcal{L}^{-1}\left\{\frac{n!}{s^{n+1}}\right\} = t^n$
Sine	$\mathcal{L}^{-1}\left\{\frac{a}{s^2 + a^2}\right\} = \sin(at)$
Cosine	$\mathcal{L}^{-1}\left\{\frac{s}{s^2 + a^2}\right\} = \cos(at)$
Transfer function	$G(s) = \frac{C(s)}{R(s)} \Big _{\text{zero I.C.}}$
Output from TF	$C(s) = R(s) G(s)$

7. MATLAB Companion

7.1. Preliminary Steps Before Running Any Code

The following steps **must be completed** before executing any MATLAB command in this section.

Step 1. Verify the Symbolic Math Toolbox is installed.

In the Command Window type:

```
>> ver
```

Look for Symbolic Math Toolbox in the list. If missing, install via: *Home* → *Add-Ons* → *Get Add-Ons*.

Step 2. Clear the workspace before starting.

Always begin with a clean environment:

```
>> clear; clc; close all;
```

This removes any variables, command history, and figures from a previous session that could conflict with the current one.

Step 3. Declare symbolic variables once at the top.

```
>> syms t s a b c n real
```

Do not redeclare inside loops or functions. The variables **t** and **s** are the time and complex-frequency variables used in every problem.

Step 4. Understand the two core commands.

MATLAB's Symbolic Math Toolbox provides:

- $F = \text{laplace}(f, t, s)$ — forward transform: $f(t) \rightarrow F(s)$
- $f = \text{ilaplace}(F, s, t)$ — inverse transform: $F(s) \rightarrow f(t)$

Both assume $t \geq 0$ (one-sided transform with lower limit 0^-).

Step 5. Simplify every result.

Raw output from `laplace` / `ilaplace` can look messy. Always apply:

```
>> F = simplify(laplace(f, t, s))
```

Step 6. Run section by section.

Each block below is preceded by `%`. Use *Ctrl + Enter* (Run Section) to execute one block at a time. This makes it easier to debug and verify intermediate results.

Step 7. Suppress unwanted output.

Add a semicolon ; at the end of a line to suppress printing. Remove it to see the intermediate value in the Command Window.

Step 8. Interpret the output.

MATLAB may display results differently from the textbook but they are algebraically equivalent. Always use `simplify()` or `simplifyFraction()` to compare with hand-calculated results.

7.2. Part 1 — Forward Laplace Transforms**7.2.1. Setup**

```
clear; clc; close all;
syms t s a b c n real
```

7.2.2. Example 2.1 — $\mathcal{L}\{Ae^{-at}u(t)\} = A/(s + a)$

```
syms A positive
f = A * exp(-a*t);
F = simplify(laplace(f, t, s))
% Result: A/(s+a)
```

7.2.3. Derivation — $\mathcal{L}\{e^{at}\} = 1/(s - a)$

```
f = exp(a*t);
F = simplify(laplace(f, t, s))
% Result: 1/(s-a)
```

7.2.4. Example A — $g(t) = 4\cos(4t) - 9\sin(4t) + 2\cos(10t)$

```
g = 4*cos(4*t) - 9*sin(4*t) + 2*cos(10*t);
G = simplify(laplace(g, t, s))
% Result: 4s/(s^2+16) - 36/(s^2+16) + 2s/(s^2+100)
```

7.2.5. Example B — $h(t) = 3\sinh(2t) + 3\sin(2t)$

```
h = 3*sinh(2*t) + 3*sin(2*t);
H = simplify(laplace(h, t, s))
% Result: 6/(s^2-4) + 6/(s^2+4)
```

7.2.6. Example C — $g(t) = e^{3t} + \cos(6t) - e^{3t}\cos(6t)$

```
g = exp(3*t) + cos(6*t) - exp(3*t)*cos(6*t);
G = simplify(laplace(g, t, s))
% Result: 1/(s-3) + s/(s^2+36) - (s-3)/((s-3)^2+36)
```

7.2.7. Example D — $f(t) = t \cosh(3t)$ (Entry #30)

```
f = t * cosh(3*t);
F = simplify(laplace(f, t, s))
% Result: (s^2+9)/(s^2-9)^2

% Manual verification via entry #30: F = -d/ds[L{cosh(3t)}]
G_cosh = s/(s^2 - 9);
F_check = simplify(-diff(G_cosh, s))
```

7.2.8. Example E — $h(t) = t^2 \sin(2t)$ (Entry #30 twice)

```
h = t^2 * sin(2*t);
H = simplify(laplace(h, t, s))
% Result: (12s^2-16)/(s^2+4)^3

% Verify: apply entry #30 to F(s) = 4s/(s^2+4)^2 [entry #9]
F_tsin = 4*s / (s^2+4)^2;
H_check = simplify(-diff(F_tsin, s))
```

7.2.9. Example F — $g(t) = t^{3/2}$ (Entries #5 and #32)

```
g = t^(sym(3)/2);
G = simplify(laplace(g, t, s))
% Result: 3*sqrt(pi) / (4*s^(5/2))
```

7.2.10. Example G — $f(t) = (10t)^{3/2}$ (Entry #24 time scaling)

```
f = (10*t)^(sym(3)/2);
F = simplify(laplace(f, t, s))

% Verify via scaling: F(s) = (1/10)*G(s/10)
G_F = sym('3*sqrt(pi)/(4*s^(5/2))');
F_check = simplify(subs(G_F, s, s/10)/10)
```

7.2.11. Linearity Example — $f(t) = 6e^{-5t} + e^{3t} + 5t^3 - 9$

```
f = 6*exp(-5*t) + exp(3*t) + 5*t^3 - 9;
F = simplify(laplace(f, t, s))
% Result: 6/(s+5) + 1/(s-3) + 30/s^4 - 9/s
```

7.3. Part 2 — Inverse Laplace Transforms**7.3.1. Example (a) — $F(s) = 6/s - 1/(s-8) + 4/(s-3)$**

```
F = 6/s - 1/(s-8) + 4/(s-3);
f = simplify(ilaplace(F, s, t))
% Result: 6 - exp(8t) + 4*exp(3t)
```

7.3.2. Example (b) — $H(s) = 19/(s+2) - 1/(3s-5) + 7/s^5$

```
H = 19/(s+2) - 1/(3*s-5) + 7/s^5;
h = simplify(ilaplace(H, s, t))
% Result: 19*exp(-2t) - (1/3)*exp(5t/3) + (7/24)*t^4
```

7.3.3. Example (c) — $F(s) = 6s/(s^2+25) + 3/(s^2+25)$

```
F = 6*s/(s^2+25) + 3/(s^2+25);
f = simplify(ilaplace(F, s, t))
% Result: 6*cos(5t) + (3/5)*sin(5t)
```

7.3.4. Example (d) — $G(s) = 8/(3s^2+12) + 3/(s^2-49)$

```
G = 8/(3*s^2+12) + 3/(s^2-49);
g = simplify(ilaplace(G, s, t))
% Result: (4/3)*sin(2t) + (3/7)*sinh(7t)
```

7.4. Part 2 — Partial Fractions**7.4.1. PF Example (a) — Three Linear Factors**

$$G(s) = \frac{86s - 78}{(s+3)(s-4)(5s-1)}$$

```
G = (86*s - 78) / ((s+3)*(s-4)*(5*s-1));
g = simplify(ilaplace(G, s, t))
% Result: -3*exp(-3t) + 2*exp(4t) + exp(t/5)
```

7.4.2. PF Example (b) — Linear + Irreducible Quadratic

$$F(s) = \frac{2 - 5s}{(s-6)(s^2+11)}$$

```
F = (2 - 5*s) / ((s-6)*(s^2+11));
f = simplify(ilaplace(F, s, t))
% Result: -(28/47)*exp(6t) + (28/47)*cos(sqrt(11)*t)
%          - (67/(47*sqrt(11)))*sin(sqrt(11)*t)
```

7.4.3. PF Example (c) — Higher-Order Denominator

$$G(s) = \frac{25}{s^3(s^2+4s+5)}$$

```
G = 25 / (s^3 * (s^2 + 4*s + 5));
g = simplify(ilaplace(G, s, t))
```

7.5. Part 2 — Transfer Functions and System Response

7.5.1. Example 2.5 — Unit Step Response, $G(s) = 1/(s + 2)$

```
G = 1/(s+2);
R = 1/s;                                     % unit step: r(t)=u(t)
C = simplify(R * G);
c = simplify(ilaplace(C, s, t))
% Result: 1/2 - (1/2)*exp(-2t)

% Plot
t_vec = 0:0.01:3;
c_num = double(subs(c, t, t_vec));
plot(t_vec, c_num, 'b-', 'LineWidth', 2)
xlabel('t (s)'), ylabel('c(t)')
title('Step Response: G(s)=1/(s+2)'), grid on
```

7.5.2. Skill-Assessment 2.3 — Transfer Function from ODE

The ODE: $\frac{d^3c}{dt^3} + 3\frac{d^2c}{dt^2} + 7\frac{dc}{dt} + 5c = \frac{d^2r}{dt^2} + 4\frac{dr}{dt} + 3r$

```
num = [1, 4, 3];          % s^2 + 4s + 3
den = [1, 3, 7, 5];       % s^3 + 3s^2 + 7s + 5
G = tf(num, den)
% Result: G(s) = (s^2+4s+3)/(s^3+3s^2+7s+5)
```

7.5.3. Skill-Assessment 2.4 — ODE from Transfer Function

$G(s) = (2s + 1)/(s^2 + 6s + 2)$

```
% Cross-multiply: (s^2+6s+2)*C(s) = (2s+1)*R(s)
% Inverse Laplace: c'' + 6c' + 2c = 2r' + r
G = tf([2,1],[1,6,2]);
impz(G) % plot impulse response to verify
```

7.5.4. Skill-Assessment 2.5 — Ramp Response

$G(s) = s/[(s + 4)(s + 8)], \quad r(t) = t \Rightarrow R(s) = 1/s^2$

```
G = s / ((s+4)*(s+8));
R = 1/s^2; % ramp input
C = simplify(R * G);
c = simplify(ilaplace(C, s, t))
% Result: 1/32 - (1/16)*exp(-4t) + (1/32)*exp(-8t)
```

7.5.5. Bonus: Verify Key Table Entries

```
syms t s a n real
```

```
entries = {sin(a*t), cos(a*t), sinh(a*t), cosh(a*t), ...  
           exp(a*t), t^n, sym(1)};  
for k = 1:numel(entries)  
    F = simplify(laplace(entries{k}, t, s));  
    fprintf('L{%s} = %s\n', char(entries{k}), char(F))  
end
```